Scenario -> File Upload

❌ Weak: "Generate test cases for file upload."

Generating comprehensive test cases for a file upload feature requires covering positive scenarios, negative scenarios, security, performance, and usability.

Here is a structured set of test cases organized by category:

1. Functional Test Cases (Positive Scenarios)

Verify that the feature works as expected under normal conditions.

- **Valid File Upload:** Upload a file with a supported format and size (e.g., .jpg, .png, .pdf under the maximum size limit) and verify it uploads successfully and displays a success message.
- **Upload via Drag and Drop:** Drag and drop a valid file into the upload drop zone and verify it is accepted and queued/uploaded correctly.
- **Upload via File Browser:** Click the "Browse" or "Upload" button, select a valid file from the local file system, and verify it uploads successfully.
- **Multiple File Upload (if supported):** Select and upload multiple valid files simultaneously and verify all of them upload successfully.

- **File Replacement/Overwriting:** Upload a file with the same name as an existing uploaded file and verify the system handles it correctly (either overwrites, renames automatically, or prompts the user).

- **Cancel Upload:** Start uploading a large valid file and click the "Cancel" button mid-upload; verify the upload stops and partial data is removed from the server.

2. Validation & Negative Test Cases

Verify how the system handles invalid inputs and edge cases.

- **Invalid File Format/Extension:** Attempt to upload an unsupported file type (e.g., .exe, .bat, .sh when only images are allowed) and verify the system rejects it with an appropriate error message.

- **File Size Exceeding Limit:** Attempt to upload a file larger than the defined maximum size limit (e.g., uploading a 15MB file when the limit is 10MB) and verify it is rejected with a clear error message.

- **Zero-Byte (Empty) File:** Attempt to upload an empty file (0 KB) and verify how the system handles it (rejection or specific handling).

- **Special Characters in Filename:** Upload a file with special characters, spaces, or non-English/Unicode characters in the filename and verify the system either processes it safely or sanitizes the filename correctly.

- **Extremely Long Filename:** Upload a file with a filename exceeding the maximum character limit and verify the system handles it gracefully.

- **No File Selected:** Click the submit/upload button without selecting any file and verify a validation prompt appears.

3. Security Test Cases

Ensure the file upload mechanism is secure against malicious attacks.

- **Malicious File / Malware Upload:** Upload a test file containing a harmless script or mock virus (like the EICAR standard anti-virus test file) to ensure the system or integrated antivirus scans and blocks it.

- **MIME-Type Spoofing:** Change the extension of a dangerous executable file (e.g., malware.exe) to a safe extension (e.g., malware.jpg) and attempt to upload it. Verify that the system inspects the file header/content rather than relying solely on the file extension.

- **Path Traversal Attack:** Attempt to upload a file with a manipulated filename containing path traversal sequences (e.g., ../../../etc/passwd or ..\file.txt) and verify the system neutralizes or rejects the filename.

- **Remote Code Execution (RCE):** Ensure that uploaded files (especially scripts like .php, .asp, .jsp) cannot be directly executed from the upload directory on the server.

- **Authentication and Authorization:** Attempt to upload a file without being logged in, or as a user without the proper permissions, and verify access is denied (401/403 status).

4. Performance & Boundary Test Cases

Test the system under load and at its operational limits.

- **Maximum Allowed Size Boundary:** Upload a file that is *exactly* at the maximum size limit and one byte *over* the limit to verify boundary conditions.

- **Simultaneous Users Uploading:** Simulate multiple users uploading files at the same time to test server load, bandwidth management, and queue stability.

- **Large File Upload Performance:** Measure the time taken to upload a maximum-size file and verify that a progress bar or loading indicator updates smoothly.

5. UI/UX Test Cases

Verify the user interface and overall experience.

- **UI Elements Presence:** Verify that the upload button, drop zone, file format restrictions, and size limit guidelines are clearly visible to the user.

- **Progress Indicator:** Verify that a progress bar, percentage indicator, or spinner appears during the upload process.

- **Error Message Clarity:** Ensure that error messages are user-friendly, descriptive, and clearly state *why* an upload failed (e.g., "File size exceeds the 5MB limit" instead of a generic "Error 500").

- **Post-Upload Preview:** If applicable, verify that a thumbnail or preview of the uploaded file is correctly displayed after a successful upload.

- **Remove/Delete Uploaded File:** Verify that users can remove a successfully uploaded file before final form submission.

Using ICEPOT method: ->

I:-> Write Test Cases to test the file upload functionality

C: -> There are files of different formats like xlsx, doc, text, csv, etc.

E: -> Consider example like -> Upload the files of different sizes and formats, remove the uploaded file, no file size limit

P: -> Act as a Test Engineer

O: -> Provide output in the form of a table with proper headers like, Test Case ID, Pre-condition, Test Case name, Description, Steps, Expected Result, etc.

T: -> Test Cases should have the maximum coverage

O/P using above Prompt ->

Test Cases for File Upload

Export

TC_UP_02	User is logged in and on the file upload page.	Upload Valid Word Document (.doc / .docx)	Verify that a valid Word document uploads successfully.	1. Select a valid `.doc` or `.docx` file. 2. Confirm the selection and upload.	File should upload successfully and appear in the uploaded files list.
TC_UP_03	User is logged in and on the file upload page.	Upload Valid Text File (.txt)	Verify that a plain text file uploads successfully.	1. Select a valid `.txt` file. 2. Initiate the upload.	File should upload successfully and display properly.
TC_UP_04	User is logged in and on the file upload page.	Upload Valid CSV File (.csv)	Verify that a valid comma-separated values file uploads successfully.	1. Select a valid `.csv` file. 2. Click upload.	File should upload successfully without parsing or format errors.
TC_UP_05	User is logged in and on the file upload page.	Upload Multiple Files of Different Formats	Verify that multiple files of mixed formats (`.xlsx` , `.doc` , `.txt` , `.csv`) can be selected and uploaded simultaneously.	1. Select multiple files of different formats (e.g., sample.xlsx, notes.txt, data.csv). 2. Upload them together.	All selected files should upload successfully and be displayed concurrently.
TC_UP_06	User is logged in and on the file upload page.	Upload Extremely Large File (No File Size Limit context)	Verify the system handles exceptionally large files smoothly since there is no rigid file size restriction.	1. Select a very large file (e.g., 500MB+ `.xlsx` or `.csv`). 2. Initiate upload and monitor progress.	File should upload completely, showing a progress indicator, without timing out or crashing the application.
TC_UP_07	A file (`.xlsx` , `.doc` ,	Remove/Delete Uploaded	Verify that a user can	1. Locate the uploaded file	The file should be

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